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Comparisons of Countermovement Jump Force-Time Characteristics Among National Collegiate Athletic Association Division I American Football Athletes: Use of Principal Component Analysis

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Abstract

Merrigan, JJ, Rentz, LE, Hornsby, WG, Wagle, JP, Stone, JD, Smith, HT, Galster, SM, Joseph, M, and Hagen, JA. Comparisons of countermovement jump force-time characteristics among NCAA division I american football athletes: use of principal component analysis. *J Strength Cond Res* XX(X): 000–000, 2021—This study aimed to reduce the dimensionality of countermovement jump (CMJ) force-time characteristics and evaluate differences among positional groups (skills, hybrid, linemen, and specialists) within National Collegiate Athletic Association (NCAA) division I American football. Eighty-two football athletes performed 2 maximal effort, no arm-swing, CMJs on force plates. The average absolute and relative (e.g., power/body mass) metrics were analyzed using analysis of variance and principal component analysis procedures ($p < 0.05$). Linemen had the heaviest body mass and produced greater absolute forces than hybrid and skills but had lower propulsive abilities demonstrated by longer propulsive phase durations and greater eccentric to concentric mean force ratios. Skills and hybrid produced the most relative concentric and eccentric forces and power, as well as modified reactive strength indexes (RSI_{MOD}). Skills (46.7 ± 4.6 cm) achieved the highest jump height compared with hybrid (42.8 ± 5.5 cm), specialists (38.7 ± 4.0 cm), and linemen (34.1 ± 5.3 cm). Four principal components explained 89.5% of the variance in force-time metrics. Dimensions were described as the (a) explosive transferability to concentric power (RSI_{MOD}, concentric power, and eccentric to concentric forces) (b) powerful eccentric loading (eccentric power and velocity), (c) countermovement strategy (depth and duration), and (d) jump height and power. The many positional differences in CMJ force-time characteristics may inform strength and conditioning program designs tailored to each position and identify important explanatory metrics to routinely monitor by position. The overwhelming number of force-time metrics to select from may be reduced using principal component analysis methods, although practitioners should still consider the various metric's applicability and reliability.

Key Words: power, reactive strength index, PCA, neuromuscular performance, force plates, sport science

Introduction

Countermovement jump (CMJ) force-time data analysis is a popular method of assessing the neuromuscular status in various athletic populations (4,28,29,38). Analyses typically consider discrete parameters (e.g., jump height and peak power), the timing of characteristic events (e.g., start of the braking phase and start of the concentric phase), or waveform shape (e.g., unimodal and bimodal) (12). Moreover, variations in a movement's force-time characteristics influence the task outcome (29). Thus, the CMJ demands optimal coordination and timing of force production to maximize impulse, take-off velocity, and resultantly vertical center of mass displacement (i.e., jump height). Furthermore, the CMJ shares many common kinetic and kinematic requirements with more complex sporting actions involving the lower body, such as blocking, tackling, and route running in American football, making its application within player development very reasonable.

Owing to the diversity of inferences that can be made from force-time metrics, it is likely that certain force-producing strategies in the CMJ would be better suited for transferring to different sports or positions within the same sport (18,20). Capturing and understanding variations in jump strategy is of great interest to strength and conditioning practitioners, as this provides critical insights during athlete assessment batteries and can influence programming decisions aimed at driving higher absolute and relative power outputs on an individual and team basis (3,10). Describing the strategy and magnitude of force applications improves longitudinal performance monitoring throughout an athlete's training to identify training-related stress or adaptation. To assist in these efforts, force plate analyses allow for the quantification of an athlete's jump strategy through the calculation of various force-time characteristics. Indeed, solely capturing jump height by traditional methods, such as a Vertec or jump mat, may fail to ascertain important jump characteristics (13,29).

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The skills and functionality requirements within American football players differ among positional groups according to the biomechanical and physiological demands at each position (e.g., blocking for linemen and speed and agility for cornerbacks) (32). The variability in function is further accentuated by vast differences in anthropometrics, such as body mass, between positions (2,26,32). For example, there are vast differences among positions in vertical jump height and peak power capabilities, with the largest margins existing between skilled players and linemen (22). Considering the tactical on-field purpose of each position and aforementioned differences in anthropometrics, it is likely that movement strategies used to achieve task outcomes also differ among positions. Yet, previous literature on Division I American football players have almost exclusively measured jump height as the sole outcome measure (6,8,24,30,37), warranting further investigation of CMJ force-time characteristics within competitive American football players (e.g., National Collegiate Athletic Association [NCAA] Division 1, power 5).

Monitoring and profiling of an athlete's neuromuscular capabilities, as indicated by movement strategy or functional performances, have been widely implied through standard statistical procedures (e.g., generalized linear model) (6,17,20,22,31). Yet, the time series nature of force data and integration techniques used to calculate additional metrics results in high multicollinearity among force-time characteristics. Resultantly, many overlapping variables make similar contributions to explained variance of CMJ performances (i.e., peak velocity and peak power). However, in an attempt to enhance understanding of the complex relationships among many intercorrelated CMJ variables, a principal component analysis (PCA) can identify the strategic differences that relate to functional outcomes (7,18,28,33,38), as well as differing characteristics respective to sport-specific positional groups (35). By identifying groups of variables based on contributions to performance variation, positional differences in strategy can be recognized to inform the development of valuable performance parameters (13,14). Furthermore, identifying force-time metrics through dimensional reduction techniques allows practitioners to focus on the most relevant metrics containing different information, ultimately to yield the most optimal training adaptions over sequential training blocks. However, further research is warranted across the sports domain to inform practitioners of variables that are relevant to performances in their sport (13). Moreover, there has been recent questioning of the shortcomings of presenting PCA in sport literature (33), which warrants further meticulous investigation to strengthen the evidence of its use. The aims of this study were to (a) evaluate the CMJ force-time characteristics among positional groups within NCAA Division I American football athletes and (b) reduce the dimensionality of CMJ performance and force-time metrics.

Methods

Experimental Approach to the Problem

To evaluate the purposes of the current study, a cross-sectional research design was used on CMJ testing data during the off-season (January). Owing to the training schedule and large number of athletes, testing was conducted on 2 separate days at approximately the same time of day. Before conducting the CMJ testing, all athletes completed a brief standardized and supervised dynamic warm-up (3 sets of 20 jumping jacks, 1 set of 5 midthigh clean pull with 20 kg, and 3 sets of 5 midthigh clean pull with 40 kg).

Subjects

The study sample comprised 82 athletes (aged 18-22 years) from a division I NCAA football team. Positions within the team were categorized into linemen (offensive, $n = 12$ and defensive, $n = 9$), hybrid (running backs, $n = 6$; tight end/fullbacks, $n = 4$; and linebackers, $n = 8$), skill (wide receivers, $n = 13$; cornerbacks, $n = 8$; and safeties, $n = 9$), and specialists (quarterbacks, $n = 6$; kickers/punters, $n = 5$; and long snapper, $n = 2$). The West Virginia University Institutional Review Board (Protocol # 1807206133) approved the use of pre-existing data without requiring subject consent as a retrospective analysis of deidentified data.

Procedures

Countermovement jump Assessment. After the dynamic warm-up, athletes performed 2 warm-up CMJs (one CMJ with 50% effort and one CMJ with 75% effort) and 2 maximal effort CMJs that were recorded using ForceDecks FD4000 Dual Force Platforms (Vald Performance, Brisbane, QLD, Australia). Before the assessment, the force plates were powered on at least 45 minutes before data collection and then zeroed before each trial without the athlete standing on them. Next, the athletes were instructed to stand upright with hands grasping a polyvinyl chloride (PVC) pipe placed across their shoulders, similar in stance and positioning to a high bar back squat. The PVC pipe was used to eliminate arm swing and to ensure standardization across all athletes. The athletes remained as still as possible, while the software verified system mass (at least one still second) before performing the CMJs. The force-time curve was visually inspected to ensure limited movement occurred in system mass collection, and a signal was provided by the software when accurate body mass measure was taken. The athlete was then instructed to start, on the "3-2-1 jump!" countdown, with equal weight distribution on both force cells, drop into a self-selected countermovement depth, perform a maximal effort vertical jump "as quickly and explosively as possible," and land back onto the force plates and return to the starting position. For each jump, verbal encouragement was provided to ensure maximal effort was given during each attempt.

Force plate data were sampled at 1,000 Hz and analyzed by ForceDecks software (Vald Performance, Brisbane, QLD, Australia). This software detects the initiation of movement as a 30 N deviation from the initial system weight calculation, eccentric to concentric phase moment as the lowest center of mass displacement, and take-off as the moment the vertical forces fall 30 N below system weight. The force-time metrics were averaged across both maximal effort CMJ attempts and used for analyses. Jump height was calculated using the impulse momentum theorem and flight time methods.

Statistical Analyses

Data are reported as mean ± *SD* with respective 95% confidence intervals. One-way analyses of variance were used to evaluate the differences among positional groups. When significant univariate effects were noted, *post hoc* Tukey multiple comparisons of means *t* tests were run to determine where differences existed among positional groups.

Principal component analyses (PCAs) were conducted, using the "FactoMineR" package (function "PCA"), to reduce the dimensionality of force-time metrics and further delineate

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variations among positional groups. The correlational method of PCA was used because the force-time metrics are of different scales, which would greatly influence the results from the covariance method of PCA (13). In addition, when using the covariance method, a single variable may dominate the first principal component if the variable represents a large percentage of the total explained variance. However, the portion of total variance among positional groups explained by each metric was considered. Owing to the high collinearity among the absolute and relative value of the same metrics and power limitations, a subset of metrics was created to improve the model's validity (Table 1). This subset of data was checked for appropriate use in factor analysis by confirming sufficient correlations through Bartlett's test of sphericity (check_spericity function of "parameters" package) and sampling adequacy by the Kaiser-Meyer-Olkin test (kmo function of "psych" package) with the following cut points: <0.50, unacceptable; 0.50–0.60, miserable; 0.60–0.70, mediocre; 0.70–0.80, middling; 0.80–0.90, meritorious; and >0.90, marvelous (16). The principal components that explained a majority of the variance in jump performances were identified using a scree plot and confirmed as having eigenvalues >1.0 (15). To aid in the feature reduction process, the squared cosine ($\cos^2$), a measure of variable importance, was calculated for each metric, and the v-test was reported as a normal distribution criterion for positional group comparisons (1). Larger $\cos^2$ values indicate a larger contribution to the respective component, such that the size of the loading (in a PC biplot) is dependent on this value. The v-test score above 1.96 indicated the positional group was significantly categorized by the PC with a positive v-test indicating overexpressions (e.g., higher performances) and a negative v-test indicating underexpressions (e.g., lower performances). All statistical analyses were completed in R version 4.0.3 (36) with the critical rejection region set to a p-value < 0.05.

Results

Linemen (132.0 ± 14.5 kg) had greater body mass ($p < 0.001$) than hybrid (98.5 ± 8.68 kg), skill (86.8 ± 6.0 kg), and specialist groups (92.4 ± 6.4 kg), whereas hybrid had greater body mass than skill ($p < 0.001$) and specialists did not differ from hybrid ($p = 0.371$) or skill ($p = 0.366$). The *post hoc* results of positional differences in absolute values for overall jump performances, eccentric phase of the CMJ, and concentric phase of the CMJ are displayed in Tables 2–4, respectively. Force-time metrics that did not yield significant univariate effects among positions are as follows: countermovement depth ($p = 0.810$), eccentric to concentric peak power ratio ($p = 0.439$), braking phase duration ($p = 0.482$), eccentric peak velocity ($p = 0.073$), eccentric duration ($p = 0.186$), eccentric deceleration phase duration ($p = 0.220$) eccentric acceleration phase duration ($p = 0.361$), and eccentric to concentric duration ratio ($p = 0.101$). Furthermore, the univariate effect of position groups on eccentric deceleration rate of force development (RFD) was significant ($p = 0.048$) but yielded no differences across position groups.

Similarly, descriptive values for relative force-time metrics of the CMJ are displayed in Table 5. Relative force-time metrics that did not yield significant univariate effects among positions were eccentric braking RFD 0–100 ms ($p = 0.772$) and eccentric mean power scaled to body mass ($p = 0.081$). In addition, for a visual representation of differences across positional groups, an example force-time curve from a representative athlete within each positional group is displayed in Figures 1A and 1B.

Bartlett's test of sphericity suggested sufficient correlations among variables for factor analyses (χ^2_{351}) = 6643.17, $p < 0.001$). Furthermore, factor adequacy was considered as meritorious (overall sampling adequacy = 0.80). Four principal components (PCs) explained a cumulative 89.5% variance in CMJ performances (PC1, 52.3%; PC2, 22.2%; PC3, 9.2%; and PC4, 5.8%). Eigenvalues for principal components 1, 2, 3, and 4 were 14.1, 6.0, 2.5, and 1.6, respectively. The metrics driving each PC are listed in Table 6. Biplots were used to graphically display principal component 1 and 2, as well as 1 and 3, in Figure 2. The linemen had a significant negative loading on PC1 (v-test = −5.6). Skill players had a significant positive loading on PC 1 (v-test = 4.4) and PC2 (v-test = 2.4). Thus, in PC1, skill players and linemen had opposing loadings with skills having higher performances and linemen having lower performances, although these patterns also existed for other PCs, as indicated by the distances between the skill and linemen player groupings in Figure 1.

Discussion

The purpose of this study was to investigate positional differences in CMJ performances and force-time characteristics within a

Table 1
Force-time metrics included in the principal component analysis.*

Concentric/propulsive phase and overall performance metrics	Eccentric/braking phase and movement strategy metrics
Body mass	Countermovement depth
Modified reactive strength index	Braking duration
Jump height (impulse momentum)	Eccentric duration
Concentric duration	Eccentric peak velocity
Concentric peak velocity	†Eccentric braking RFD 0–100 ms
†Concentric mean force	†Eccentric braking RFD
†Concentric mean power	†Eccentric deceleration RFD
†Concentric peak force	†Eccentric mean power
†Concentric RPD 0–50 ms	†Eccentric peak force
†Concentric RPD 0–100 ms	†Eccentric peak power
†Total concentric RPD	Eccentric to concentric duration ratio
†Peak power	Eccentric to concentric peak power ratio
†Peak net takeoff force	Eccentric to concentric mean force ratio
†Takeoff peak force	

*ms = milliseconds; RPD = rate of power development; RFD = rate of force development.
†Force and power metrics that were scaled to individual's body mass.

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Table 2

Comparison of overall jump performances by the position group.*†

Force-time metric	Linemen (n = 21)	Specialists (n = 13)	Hybrid (n = 18)	Skills (n = 30)
Jump height (cm)	34.1 ± 5.3[c,d] (31.9, 36.3)	38.7 ± 4.0[d] (36.3, 41.1)	42.8 ± 5.5[a] (40.2, 45.4)	46.7 ± 4.6[a,b,c] (45.1, 48.4)
Jump height imp mom (cm)	32.8 ± 5.5[c,d] (30.7, 34.9)	36.3 ± 5.0[d] (33.3, 39.4)	41.5 ± 5.4[a] (39,1, 43.9)	45.3 ± 4.7[a,b] (43.4, 47.2)
Contraction time (ms)	845 ± 104[c,d] (803, 886)	814 ± 78 (758, 860)	754 ± 66[a] (723, 785)	772 ± 95[a] (737, 807)
Countermovement depth (cm)	−37.4 ± 5.9 (−40.6, −35.6)	−38.33 ± 5.3 (−40.3, −33.9)	−38.8 ± 5.0 (−41.2, −36.4)	−37.2 ± 7.0 (−39.8, −34.7)
Eccentric to concentric mean force ratio (N)	55.7 ± 4.6[b,c,d] (53.7, 57.4)	52.2 ± 3.6[a,d] (49.7, 54.1)	49.3 ± 3.4[a] (47.7, 50.9)	47.6 ± 2.8[a,b] (46.6, 48.6)
Eccentric to concentric peak power ratio (W)	0.400 ± 0.104 (0.369, 0.456)	0.420 ± 0.098 (0.349, 0.449)	0.437 ± 0.079 (0.4, 0.474)	0.385 ± 0.132 (0.337, 0.433)
Eccentric to concentric duration ratio (ms)	168 ± 16 (161, 174)	175 ± 21 (165, 190)	169 ± 12 (163, 175)	182 ± 31 (171, 193)
Modified reactive strength index (m·s−1)	0.41 ± 0.10[c,d] (0.37, 0.45)	0.48 ± 0.08[d] (0.43, 0.54)	0.57 ± 0.09[a] (0.53, 0.62)	0.61 ± 0.09[a,b] (0.58, 0.65)

*AU =.

†Data are presented as mean ± *SD* (upper, lower 95% confidence intervals) statistical significance at $p < 0.05$ is indicated by the following: a, different than linemen; b, different than specialist; c, different than hybrid; and d, different than skill.

Division I Power 5 American football team. To manage the high dimensionality of CMJ force-time data, a PCA suggested that the variance in discrete CMJ force-time metrics were explained by (a) explosive transferability of generated eccentric forces to rapid concentric power, (b) fast and powerful eccentric loading, (c) countermovement strategies of depth and duration, and (d) the overall jumping capability from rapid eccentric force development and concentric velocities. Overall, linemen produced greater absolute forces than hybrid and skills athletes but may not be enough to offset the difference in body mass to generate similar propulsive capabilities (i.e., jump heights), which can be mainly attributed to elongated concentric durations. Logically, after controlling for body mass, the hybrid and skills athletes produced forces and power at greater magnitudes and faster rates than linemen. These findings demonstrate performance abilities in absolute and relative perspectives and provide rationale for selecting variables of interest when monitoring CMJ force-time metrics in American football athletes.

The sport of American football is unique as it requires a wide variety of capabilities across positional groups within a team, which are exacerbated at higher competitive levels (32). Owing to variations in positional roles during gameplay, differences in relative and absolute force-time metrics in this study are expected. The CMJ force-time characteristics likely differ across positions as a result of innate differences in anthropometric characteristics, such as body size and composition. Although body composition or muscle physiology assessments were not included, which might provide greater context with respect to skeletal muscle characteristics and future research should consider, body mass stood out as the primary distinguishing characteristic altering various jump strategies by the position group. For example, previous research determined that linemen have significantly more lean body mass but concomitantly carry more fat mass than other football positions (2,26). Because body mass influences an individual's ability to rapidly accelerate their body (i.e., moving more or less mass against gravity) (23), differences in jump performances among football positions are likely mediated by body mass variability between positions (~6–45 kg) (19). Although all position groups require the ability to rapidly produce force, linemen must develop physical skills that permit the displacement of other individuals (i.e., opponent linemen), whereas skill positions are primarily concerned with rapidly displacing themselves (i.e., sprinting and changing direction). Hybrids require bursts of acceleration before countering the momentum of opposing players and, as such, must adequately develop both absolute and relative force production. In addition to the greater absolute and lower relative force productions, linemen exhibited a higher eccentric to concentric mean force ratio (eccentric forces were greater than concentric forces), which may suggest differences in movement strategies and

Table 3

Comparison of eccentric phase force-time characteristics by the position group.*†

Force-time metric	Linemen (n = 21)	Specialists (n = 13)	Hybrid (n = 18)	Skills (n = 30)
Acceleration phase duration (ms)	338 ± 63 (311, 364)	340 ± 47 (312, 368)	308 ± 33 (292, 323)	331 ± 71 (306, 357)
Braking impulse (N·s)	92.7 ± 23.6[b,d] (83.5, 102)	73.1 ± 18.4[a] (59.0, 79.1)	79.4 ± 18.7 (70.6, 88.3)	65.2 ± 19.2[a] (58.2, 72.2)
Braking RFD 100 ms (N·s^{-1})	5,780 ± 2,639[d] (4,575, 6712)	4,496 ± 1,347 (3,678, 5,421)	4,460 ± 1816 (3,605, 5,316)	3,782 ± 1,599[a] (3,198, 4,365)
Braking phase duration (ms)	338 ± 62 (308, 360)	319 ± 30 (301, 337)	301 ± 31 (286, 315)	320 ± 93 (286,354)
Deceleration impulse (N·s)	180 ± 36[b,d] (166, 195)	140 ± 29[a] (119, 144)	156 ± 22[d] (146, 166)	129 ± 33[a,c] (117, 140)
Deceleration phase duration (ms)	184 ± 35 (170, 199)	177 ± 28 (160, 194)	165 ± 17 (157, 173)	166 ± 46 (149, 182)
Deceleration RFD (N·s^{-1})	9,328 ± 3,100 (8,133, 10601)	7,631 ± 2004 (6076, 8,406)	9,509 ± 1856 (8,634, 10384)	8,478 ± 2,268 (7,650, 9,306)
Total eccentric duration (ms)	522 ± 84 (487, 557)	517 ± 66 (477, 557)	473 ± 42 (453, 493)	497 ± 85 (466, 528)
Braking mean force (N)	1,604 ± 195[b,c,d] (1,502, 1,667)	1,167 ± 139[a] (1,060, 1,197)	1,237 ± 132[a,d] (1,175, 1,299)	1,074 ± 106[a,c] (1,035, 1,112)
Deceleration mean force (N)	2,299 ± 326[b,c,d] (2,153, 2,416)	1734 ± 243[a,c] (1,553, 1770)	1923 ± 209[a,b,d] (1825, 2022)	1,663 ± 241[a,c] (1,575, 1751)
Total mean force (N)	1,322 ± 128[b,c,d] (1,241, 1,360)	932 ± 85[a] (870, 945)	967 ± 85[a,d] (927, 1,007)	853 ± 59[a,c] (831, 875)
Total mean power (W)	930 ± 180[b,c,d] (858, 1,002)	697 ± 134[a] (604, 706)	792 ± 110[a,d] (740, 844)	648 ± 152[a,c] (593, 704)
Total peak force (N)	3,025 ± 396[b,c,d] (2,847, 3,167)	2,254 ± 317[a,b] (2026, 2,278)	2,519 ± 267[a,b,d] (2,394, 2,645)	2,181 ± 243[a,c] (2092, 2,270)
Total peak power (W)	2,480 ± 655[b,d] (2,246, 2,777)	2038 ± 502[a] (1,666, 2,121)	2,402 ± 444 (2,193, 2,612)	1987 ± 626[a] (1759, 2,215)
Total peak velocity (m·s^{-1})	−1.34 ± 0.22 (−1.47, −1.27)	−1.46 ± 0.19 (−1.52, −1.32)	−1.59 ± 0.16 (−1.66, −1.51)	−1.48 ± 0.35 (−1.61, −1.35)

*RFD = rate of force development; RPD = rate of power development.

†Data are presented as mean ± *SD* (upper and lower 95% confidence intervals) statistical significance at $p < 0.05$ is indicated by the following: a, different than linemen; b, different than specialist; c, different than hybrid; and d, different than skills.

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Table 4

Comparison of concentric/propulsive phase force-time characteristics by the position group.*†

Force-time metric	Linemen (n = 21)	Specialists (n = 13)	Hybrid (n = 18)	Skills (n = 30)
Duration (ms)	322 ± 40[c,d] (322, 339)	296 ± 29 (275, 309)	281 ± 29[a] (267, 295)	275 ± 27[a] (265, 285)
Impulse (N•s)	336 ± 30[b,c,d] (320, 346)	253 ± 25.9[a] (235, 257)	279 ± 23.7[a,b,d] (268, 291)	259 ± 23.6[a] (250, 267)
Impulse 0–100 ms (N•s)	148 ± 24[b,c,d] (137, 156)	113 ± 17[a] (99, 118)	132 ± 14[a,b] (125, 139)	121 ± 13[a] (116, 126)
Mean force (N)	2,380 ± 232[b,c,d] (2,243, 2,450)	1790 ± 143[a,c] (1,681, 1827)	1965 ± 139[a,b,d] (1900, 2031)	1797 ± 123[a,c] (1752, 1841)
Mean power (W)	3,399 ± 417[b,c,d] (3,202, 3,544)	2,664 ± 339[a,c,d] (2,407, 2,762)	3,087 ± 307[a,b] (2,942, 3,231)	2,934 ± 278[a,b] (2,833, 3,036)
Peak force (N)	3,086 ± 325[b,c,d] (2,925, 3,195)	2,168 ± 275[a,c] (2077, 2,271)	2,518 ± 254[a,b,d] (2,399, 2,638)	2,240 ± 174[a,c] (2,177, 2,304)
Peak power (W)	6213 ± 689[b,c,d] (5,854, 6442)	4,890 ± 555[a,c] (4,485, 5,085)	5,531 ± 504[a,b] (5,293, 5,768)	5,257 ± 585[a] (5,043, 5,471)
Peak velocity ($m \cdot s^{-1}$)	2.68 ± 0.18[c,d] (2.62, 2.76)	2.81 ± 0.17[d] (2.70, 2.90)	2.96 ± 0.18[a,d] (2.88, 3.04)	3.10 ± 0.15[a,b,c] (3.03, 3.16)
RPD 0–50 ms ($W \cdot s^{-1}$)	33909 ± 8,292[b] (30724, 37326)	26782 ± 6095[a,c] (22074, 28416)	33553 ± 4,753[b] (31313, 35793)	30410 ± 5,781 (28299, 32521)
RPD 0–100 ms ($W \cdot s^{-1}$)	28305 ± 6680[b] (25270, 30842)	22513 ± 5,330[a,c,d] (18586, 25379)	27736 ± 4,534[b] (25598, 29873)	27390 ± 4,853[b] (25618, 29162)

*RFD = rate of force development; RPD = rate of power development.

†Data are presented as mean ± *SD* (upper, lower 95% confidence intervals) statistical significance at $p < 0.05$ is indicated by the following: a, different than linemen; b, different than specialist; c, different than hybrid; and d, different than skills.

capabilities of accelerating through the concentric phase such that linemen do so at a slower pace than skills or hybrids.

As such, previous literature identified vast differences in vertical jump height by position, or positional group, in subelite and elite football players. However, few studies evaluated this position differential based on movement strategies used to execute the CMJ (22,34). The PCA found principal component 1, 2, and 3 to involve RSI_{MOD}, countermovement strategies and ratio of force, power, and speed between the eccentric and concentric phases. These significant contributors are imperative in demonstrating the phase-specific differences that exist between positions in jump performance. According to the PCA in the current study, the largest differences existed between skill players and linemen. This was primarily demonstrated in PC1, which included the ability to transfer forces from the eccentric to concentric phase, reactive strength indexes, and the magnitude and rate of concentric power production. The durations, magnitude, and velocity of the countermovement (i.e., eccentric phase) did not differ across positions; however, linemen exhibited longer concentric phase durations. Resultantly, linemen demonstrated limited ability to produce enough propulsive forces to accelerate their heavier body mass at high enough rates to achieve higher relative jump heights, concentric power, and RSI_{MOD}. Collectively, the eccentric strategies were mostly consistent between positional groups, whereas the largest differences between the groups were their ability to transfer the energy and forces from the eccentric phase to produce equivalent concentric forces and power. It is notable that the peak power outputs produced by these athletes (Table 4) are beyond several findings in the literature of 3–4,000 W (19,21), which likely results from their level of play and sport (25,39). Therefore, alluding to the physical and neuromuscular capabilities of high-level (division I, power 5 conference) collegiate football players, which are essential for high performance in a collision sport (5), and other sports requiring high-power outputs (9,11).

The use of PCA in the current study demonstrated the ability to reduce a large number of metrics to improve the ability to monitor various components within jump performances. Previous literature demonstrated similar capabilities using dimensional reduction on performance monitoring metrics in other sports, which should be referenced and applied accordingly (13,20,35). Ultimately, it is reasonable to anticipate positional demands across various sports because of the inherent competitive demands in both team (e.g., American football, basketball, and rugby) and individual (e.g., track and field, distance running, and wrestling) sports. About research that is more specific to force-time metrics, temporal and force domains were identified, which were further broken down as output metrics (e.g., displacements), causal (e.g., forces), and timing (e.g., time to peak force) (13,20).

Table 5

Comparison of scaled force-time characteristics (relative to body mass) by the position group.*†

Force-time metric	Linemen (n = 21)	Specialists (n = 13)	Hybrid (n = 18)	Skills (n = 30)
Concentric mean force ($N \cdot kg^{-1}$)	17.8 ± 1.5[c,d] (17.2, 18.4)	19.0 ± 1.5[d] (18.1, 19.9)	20.0 ± 1.3[a] (19.4, 20.6)	20.7 ± 1.2[a,b] (20.3,21.2)
Concentric mean power ($W \cdot kg^{-1}$)	25.4 ± 3.4[c,d] (24.2, 27.1)	28.2 ± 3.6[c,d] (25.8, 30.4)	31.5 ± 3.3[a,b] (30.0, 33.0)	33.8 ± 2.9[a,b] (32.8, 34.9)
Concentric peak force ($N \cdot kg^{-1}$)	23.0 ± 1.9[c,d] (22.4, 24)	23.9 ± 1.7[c,d] (22.6, 24.6)	25.6 ± 1.6[a,b] (24.8, 26.4)	25.9 ± 2.0[a,b] (25.1, 26.6)
Concentric peak power ($W \cdot kg^{-1}$)	46.4 ± 5.6[c,d] (44.4, 49.2)	51.7 ± 6.2[d] (48.2, 55.6)	56.5 ± 6.2[a] (53.6, 59.4)	60.6 ± 5.8[a,b] (58.5, 62.7)
Concentric RPD 50ms ($W \cdot s^{-1} \cdot kg^{-1}$)	254 ± 67[c,d] (232, 288)	274 ± 64[c,d] (236, 313)	342 ± 45[a,b] (321, 363)	352 ± 70[a,b] (326, 377)
Concentric RPD 100ms ($W \cdot s^{-1} \cdot kg^{-1}$)	212 ± 57[c,d] (191, 237)	240 ± 72[d] (197, 283)	283 ± 46[a] (261, 305)	317 ± 63[a,b] (294, 340)
Concentric RPD ($W \cdot s^{-1} \cdot kg^{-1}$)	192 ± 49[c,d] (172, 212)	232 ± 55[d] (199, 265)	261 ± 47[a] (239, 283)	293 ± 53[a,b] (274, 313)
Eccentric braking RFD 100ms ($N \cdot s^{-1} \cdot kg^{-1}$)	42.6 ± 18.4 (34.9, 49.6)	47.2 ± 14.5 (39.6, 57.9)	45.7 ± 19.1 (36.7, 54.7)	43.6 ± 18.5 (36.8, 50.4)
Eccentric braking RFD ($N \cdot s^{-1} \cdot kg^{-1}$)	59.5 ± 18.3[c,d] (53.4, 68.3)	68.7 ± 12.0 (59.8, 75.1)	81.2 ± 12.6[a] (75.2, 87.1)	76.9 ± 19.4[a] (69.8, 84.0)
Eccentric deceleration RFD ($N \cdot s^{-1} \cdot kg^{-1}$)	69.7 ± 23.8[c,d] (61.7, 81.2)	80.2 ± 20.2 (65.4, 91.5)	96.8 ± 17.6[a] (88.5, 105.0)	97.9 ± 26.1[a] (88.4, 107.0)
Eccentric mean power ($W \cdot kg^{-1}$)	6.92 ± 1.20 (6.54, 7.58)	7.30 ± 0.88 (6.64, 7.51)	8.05 ± 0.88 (7.64, 8.47)	7.46 ± 1.62 (6.86, 8.05)
Eccentric peak force ($N \cdot kg^{-1}$)	22.5 ± 2.6[c,d] (21.7, 23.9)	23.7 ± 1.9 (22.2, 24.4)	25.6 ± 1.8[a] (24.8, 26.4)	25.1 ± 2.5[a] (24.2, 26.0)
Eccentric peak power ($W \cdot kg^{-1}$)	18.4 ± 4.6[c] (17.1, 21.1)	21.3 ± 3.8 (18.3, 22.5)	24.4 ± 4.1[a] (22.5, 26.3)	22.9 ± 6.9 (20.3, 25.4)
Take-off force (N)	23.1 ± 1.9 (22.5, 24.1)	24.0 ± 1.7 (22.5, 24.8)	25.7 ± 1.7 (24.8, 26.6)	26.0 ± 2.0 (25.3, 26.7)

*RFD, rate of force development; RPD, rate of power development.

†Data are presented as mean ± *SD* (upper, lower 95% confidence intervals) statistical significance at $p < 0.05$ is indicated by the following: a, different than linemen; b, different than specialist; c, different than hybrid; and d, different than skills.

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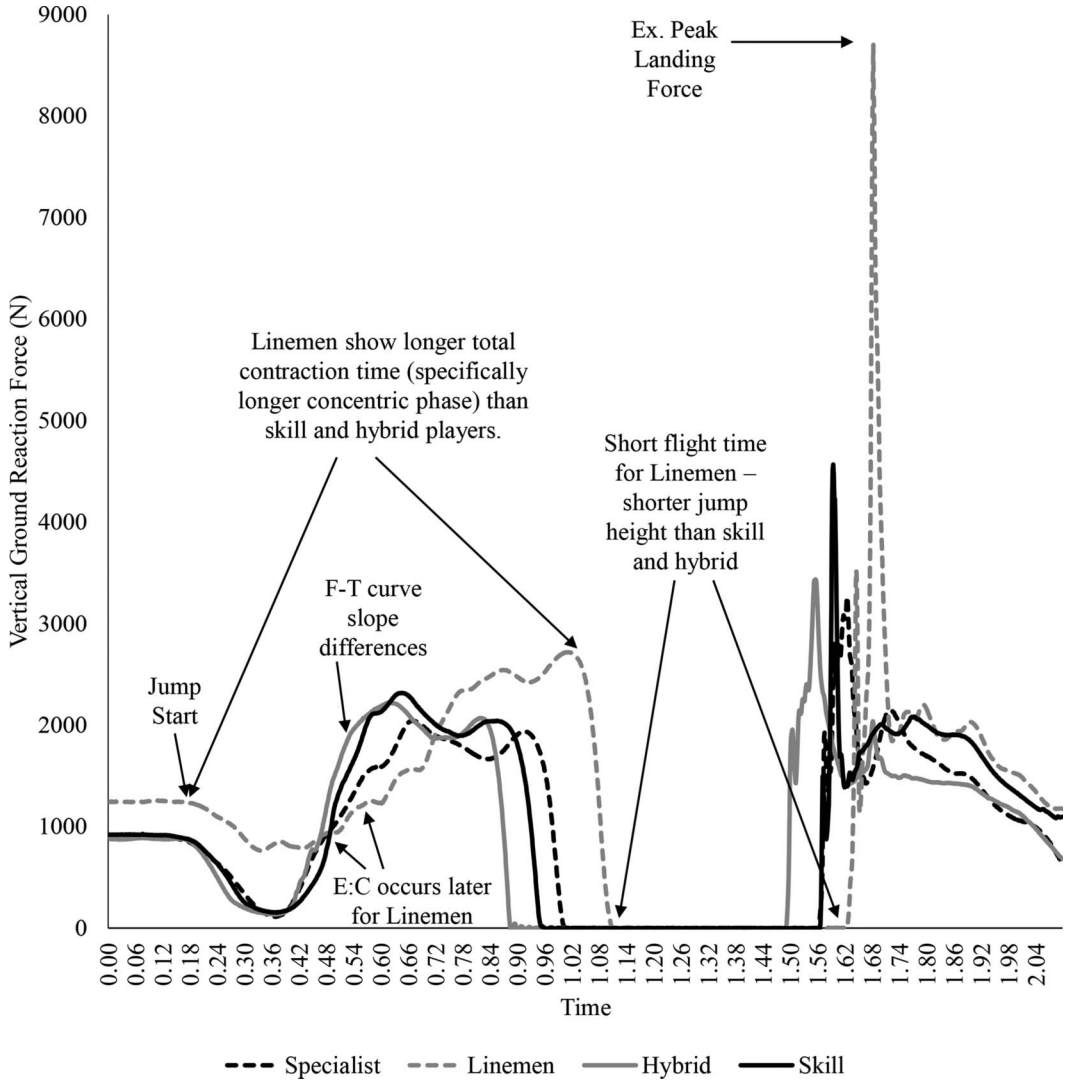

Figure 1. Visual representation of example force-time (F–T) curves from countermovement jumps of one representative athlete within each positional group (specialist, linemen, hybrid, and skills). Each athlete was selected by having an average jump height for their respective group. Several differences among positions, namely, linemen compared with skills and hybrid, are made throughout the figure. Such as, a later transition from eccentric to concentric phases (E:C), more gradual force-time slopes, longer total contractions times, and shorter flight times (lower jump heights).

Similarities occurred in the current principal component analysis, but differences in variables contributing the most to jump performances will exist depending on the sport of interest, as well as the metrics included in the analyses. Although eccentric rate of force development was included in the fourth PC, this is likely due to its relation to jump height (17). However, the reliability of rate of force development metrics must be understood to make appropriate decisions based on this metric (27) and may not provide further benefit than monitoring jump height alone. Likewise, because of the impulse momentum derivation of jump height, peak velocity and power will strongly coincide with variations in jump height (21). Therefore, we suggest selecting 1–2 metrics to monitor from each PC including explosive transferability to concentric power, powerful eccentric loading, countermovement strategies, and jump height (Table 6).

Although the differences between positions in jump height may be tested with a simpler method such as the Vertec, the recent advances in force plate software allow efficient testing with timely analysis and reporting of information beyond jump height alone. However, the most important metrics to monitor are still uncertain, leaving practitioners with the burden of selecting from numerous metrics. Often, the metrics selected will be those most related with jump height. This train of thought is sound, but often leads to monitoring metrics such as peak power or velocity and concentric impulse. Because these metrics are all derivatives of jump height, the falsely inflated relationship does not demonstrate added value to monitoring programs. Conversely, as demonstrated in the current study, force plates may be used to compare jumping strategies and abilities beyond jump height or power output. These results and suggestions are not without limitations but represent initial steps for down selecting force-time metrics for a given scenario. Overall, the PCA highlights groupings of metrics that explain the most variance in a single data set. This strategy comes with difficult interpretations to understand and identify the most important metrics within each PC (group of similar metrics). In addition, these combinations of

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Table 6

Top 5 contributing force-time metrics to the 4 principal components of force-time metrics.*†

Principal component 1		Principal component 2		Principal component 3		Principal component 4	
Explosive transferability to concentric power		**Powerful eccentric loading**		**Countermovement strategy**		**Jump height and power output**	
RSI_{MOD}	0.85	Ecc:Con power	0.90	Counter depth	0.61	Deceleration RFD	0.14
Con mean power	0.84	Ecc peak velocity	0.63	Braking duration	0.51	Con peak velocity	0.11
Con RPD 50ms	0.83	Ecc:Con duration	0.59	Ecc duration	0.48	Braking RFD100	0.11
Con RPD 100ms	0.83	Ecc mean power	0.56	Braking RFD100	0.24	Jump height	0.11
Ecc:Con force	0.82	Ecc peak power	0.55	Ecc mean power	0.14	Peak power	0.10

*RSI_{MOD} = modified reactive strength index; Concentric = Con; Eccentric = Ecc; Eccentric to Concentric phase ratio = Ecc:Con; RFD100 = rate of force development from 0 to 100 ms; RFD = rate of force development; RPD = rate of power development.
†Values are cos^2.

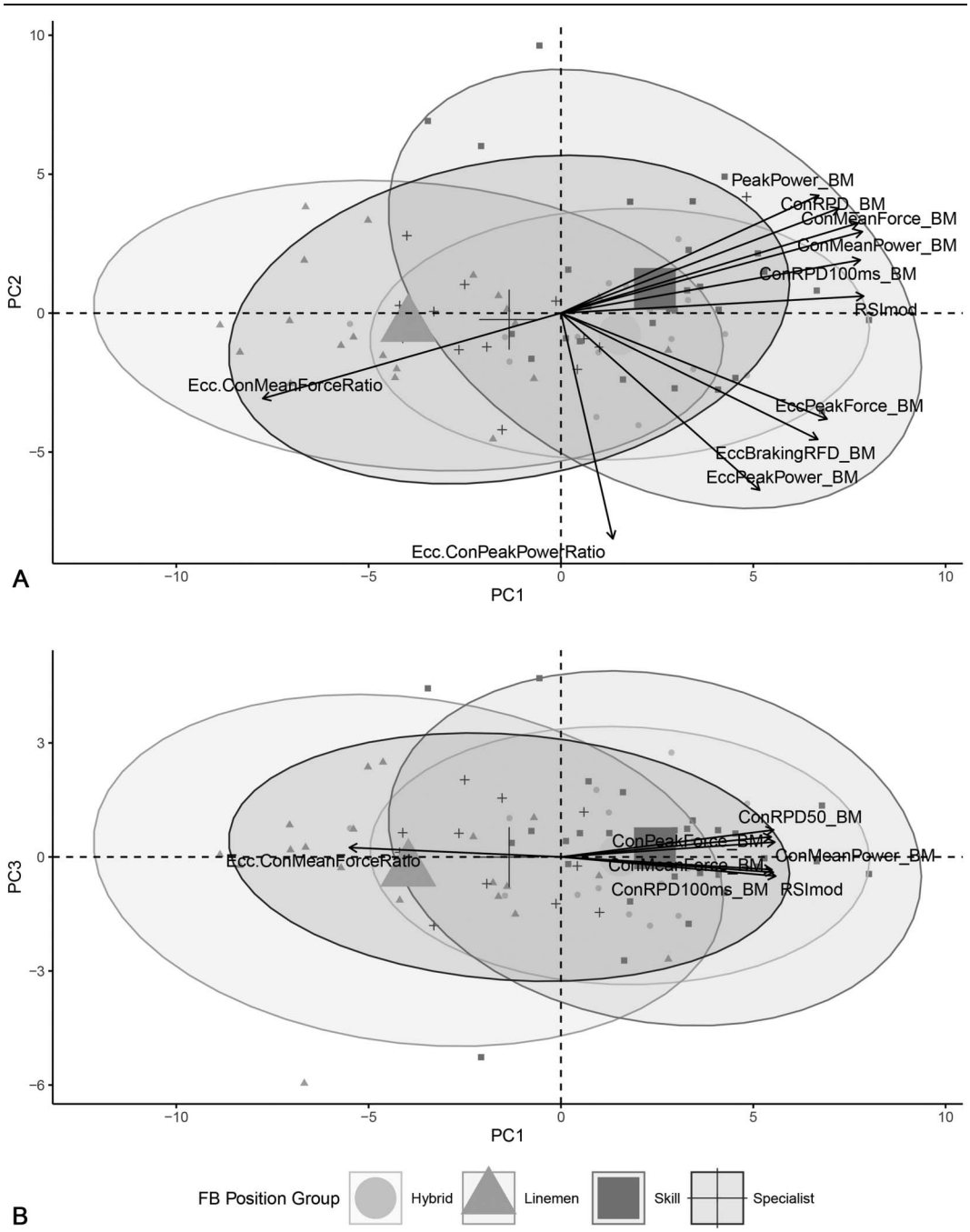

Figure 2. Loadings of principal components 1 and 2 (A) and principal components 1 and 3 (B) of force-time metrics from eccentric (Ecc) and concentric (Con) phases of the jump. All force and power metrics are scaled to body mass (BM).

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metrics formulating each PC are likely to change with different circumstances. Therefore, additional research is warranted that investigates multiple timepoints of testing, rather than one timepoint during the off-season, to understand which metrics may be most susceptible to change within this population. However, the current findings are useful for describing metrics that may vary among groups of individuals that differ in body size, as well as playing demand and intent. Owing to the aforementioned limitations of PCA, additional research and careful application are advised to ensure the application of monitoring the suggested metrics prove beneficial.

Practical Applications

The findings of the current study aid in demonstrating the innate physical characteristics and training adaptations within American football, at the division I power 5 collegiate level of competition. Although linemen are most likely to achieve the highest absolute force and power, differences in movement strategies and body mass, and subsequently reduced stretch reflex and elastic properties, are likely main contributors to differences in propulsive abilities across positional groups. As such, countermovement strategies, such as countermovement depth and eccentric velocities and power, were generally consistent between positional groups, whereas the largest differences resided in their ability to produce equivalent or greater concentric forces and power outputs. Along with the body size, these movement strategies and propulsive abilities may provide insight into the appropriate positional assignments and training program prescriptions, while also informing the metrics of interest for monitoring purposes for each positional group.

Of course, foundational levels of strength and power are crucial to a well-periodized strength and conditioning regimen for an American football athlete, given that both are influential to success during competition regardless of the position. As such, practitioners rightfully aim to maximize the development of both strength and power longitudinally. However, given the findings contained herein, some slight modifications to position-specific strength and power training might consider the variables identified by the PCA as indices for the most warranted desirable training adaptation. For example, provided linemen struggled with transferring force and power from eccentric to concentric phases, a focus of exercises during certain training blocks that purport to enhance elastic properties at the hips, knees, and ankles might aid them in improving their eccentric to concentric ratios and systemic power output. Conversely, skills and hybrids were able to displace their mass (particularly during the concentric phase) much more rapidly so it might be inferred that their elastic properties are not their weakness, rather their absolute force outputs (i.e., maximal strength) are setting the ceiling on their total power output (albeit still relatively high compared with most other populations).

The reduction in number of force-time metrics may be achieved by conducting a PCA, but practitioners should use their best judgment to select the most applicable and reliable metrics from each dimension presented by these statistical procedures, and consider the multicollinearity of different variables within a large data set such as those provided by typical force plate testing software. Since the results of these analyses may differ according to the scenario (baseline versus monitoring data), sport, and competition level, future research should continue to explore the use of these methodological procedures to inform practitioners of valuable CMJ force-time metrics for their specific sport.

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